WORKSHEET

Simple interest 1

Express all monetary answers correct to the nearest cent. Use 1 year = 52 weeks = 365 days.

1 How many months in:

a 6 years? _____ **b** 9 years? _____ **c** $2\frac{1}{2}$ years? _____ **d** $3\frac{1}{4}$ years? _____

2 How many weeks in:

a 5 years? _____ **b** 8 years? _____ **c** $4\frac{1}{2}$ years? _____ **d** $2\frac{1}{4}$ years? _____

3 Write as decimals.

a 6% _____ **b** 8.5% _____ **c** 12.25% _____ **d** 2.5% _____

e 20% _____ **f** 5.05% _____ **g** $3\frac{1}{2}$% _____ **h** $7\frac{1}{4}$% _____

4 Convert these annual percentage rates to monthly interest rates expressed as decimals.

a 13.2% p.a. _____ **b** 21% p.a. _____ **c** 8.25% p.a. _____ **d** 10.68% p.a. _____

5 Convert these rates to daily interest rates expressed as decimals correct to four significant figures.

a 16% p.a. _____ **b** 9.5% p.a. _____ **c** 22.8% p.a. _____ **d** 13.21% p.a. _____

6 **Simple interest formula** $I = Prn$

Calculate the simple interest earned when:

a $4500 is invested at 7.5% p.a. for 5 years _____

b $11 000 is invested at 5.75% p.a. for $2\frac{1}{2}$ years _____

c $6800 is invested at 3.8% p.a. for 4 years _____

d $38 000 is invested at 10% p.a. for 9 months _____

e $25 400 is invested at 6.5% p.a. for 100 days _____

f $18 600 is invested at 0.73% per month for 1 year _____

7 Calculate the time needed to invest:

a $5000 at 7% p.a. to earn $2100 interest _____

b $16 700 at 4.8% p.a. to earn $2004 interest _____

c $7800 at 3.1% p.a. to earn $181.35 interest _____

d $28 950 at 1.25% per month to earn $3618.75 interest _____

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8 Calculate the principal to be invested at:

a 11% p.a. for 4 years to earn $3036 interest _____

b $9\frac{1}{2}$% p.a. for 2.5 years to earn $2541.25 interest _____

c 5.35% p.a. for $1\frac{1}{4}$ years to earn $1471.25 interest _____

d 15% p.a. for 9 months to earn $3881.25 interest _____

9 Calculate the simple interest rate (% p.a.) if:

a $2300 invested for 6 years earns $690 interest _____

b $14 500 invested for $3\frac{1}{2}$ years earns $2283.75 interest _____

c $26 000 invested for 2 years earns $5824 interest _____

d $40 000 invested for 18 months earns $4950 interest _____

10 Final amount = principal + interest

Calculate the final amount when:

a $3200 is invested at 4% p.a. for 2.5 years _____

b $9700 is invested at 13.1% p.a. for 11 months _____

c $29 000 is invested at 9.4% p.a. for 40 weeks _____

d $21 375 is invested at 8.7% p.a. for 200 days _____

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Answers

1	**a** 72	**b** 108	**c** 30	**d** 39
2	**a** 260	**b** 416	**c** 234	**d** 117
3	**a** 0.06	**b** 0.085	**c** 0.1225	**d** 0.025
	e 0.2	**f** 0.0505	**g** 0.035	**h** 0.0725
4	**a** 0.011	**b** 0.0175	**c** 0.006 875	**d** 0.0089
5	**a** 0.000 438 4	**b** 0.000 260 3	**c** 0.000 624 7	**d** 0.000 361 9
6	**a** \$1687.50	**b** \$1581.25	**c** \$1033.60	**d** \$2850
	e \$452.33	**f** \$1629.36		
7	**a** 6 years	**b** $2\frac{1}{2}$ years	**c** $\frac{3}{4}$ year = 9 months	**d** 10 months
8	**a** \$6900	**b** \$10 700	**c** \$22 000	**d** \$34 500
9	**a** 5%	**b** 4.5%	**c** 11.2%	**d** 8.25%
10	**a** \$3520	**b** \$10 864.81	**c** \$31 096.92	**d** \$22 393.97

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